Machine Learning and Computational Statistics
Homework 3

Ex. 1 : 2
Ex. 2 : 3 + 4
Ex. 3 : 1 + 3
Ex. 4 : 2
Ex. 5 : 4

Ex. 6.

Case A : (a) $MSE = a + 0 = a$
(b) unbiased ($E(\hat{\vartheta}) = \vartheta_0$)

Case B : (a) $MSE = a + \frac{a}{3} = \frac{4a}{3}$
(b) biased

Case C : (a) $MSE = 3a + 0 = 3a$
(b) unbiased

Case D : (a) $MSE = 3a + \frac{a}{3} = \frac{10a}{3}$
(b) biased

Case E : (a) $MSE = a + 3a = 4a$
(b) biased

Ex. 7 : 3
Ex. 8 : 1 + 4
Ex. 9 : 2
Ex. 10 : 2
Ex. 11 : 4

Ex. 12.

In the regression framework consider the data set $X=\{(y_i,x_i)\ ,x_i\in\mathbb{R}^l, y_i\in\mathbb{R}, i=1,\dots,N\}$.
The aim of the ridge regression is to minimize the cost function:

$$J(\vartheta)=\sum_{n=1}^{N}(y_n-\vartheta^T x_n)^2$$

Subject to the constraint $\|\vartheta\|^2\le\rho$ where $\rho(>0)$ is a specific constant.
One way to achieve this is to define the associated Lagrangian function:

$$L(\vartheta)=\sum_{n=1}^{N}(y_n-\vartheta^T x_n)^2+\lambda\|\vartheta\|^2$$

and to minimize it with respect to ϑ.

(a) Show that the minimizer $\hat{\theta}$ of $L(\vartheta)$ satisfies the equation:

$$\left(\sum_{n=1}^{N}x_n x_n^T+\lambda I\right)\hat{\theta}=\sum_{n=1}^{N}y_n x_n$$

(b) Prove that the above solution can be expressed in matrix form as:

$$\hat{\theta}=(X^TX+\lambda I)^{-1}X^Ty$$

where $X=\begin{bmatrix}x_1^T\\ \vdots\\ x_N^T\end{bmatrix}$ and $y=\begin{bmatrix}y_1\\ \vdots\\ y_N\end{bmatrix}$

(a) $\dfrac{\partial L(\vartheta)}{\partial\vartheta}=0 \Rightarrow \dfrac{\partial}{\partial\vartheta}\sum_{n=1}^{N}(y_n-\vartheta^Tx_n)^2+\dfrac{\partial}{\partial\vartheta}(\lambda\|\vartheta\|^2)=0$

$$\Rightarrow \sum_{n=1}^{N}2(y_n-\vartheta^Tx_n)(-x_n)+2\lambda\vartheta=0$$

$$\Rightarrow -2\sum_{n=1}^{N}y_nx_n+2\sum_{n=1}^{N}(\vartheta^Tx_n)x_n+2\lambda\vartheta=0$$

$\left((\vartheta^Tx)x=(xx^T)\vartheta\right)$

$$\Rightarrow \left(\sum_{n=1}^{N}x_nx_n^T\right)\vartheta+\lambda I\vartheta=\sum_{n=1}^{N}y_nx_n$$

$$\Rightarrow \left(\sum_{n=1}^{N}x_nx_n^T+\lambda I\right)\hat{\theta}=\sum_{n=1}^{N}y_nx_n$$

(b) $\sum_{n=1}^{N}x_nx_n^T=X^TX$

$\sum_{n=1}^{N}y_nx_n=X^Ty$

(a) $\Rightarrow (X^TX+\lambda I)\hat{\theta}=X^Ty$

$\Rightarrow \hat{\theta}=(X^TX+\lambda I)^{-1}X^Ty$

Ex. 13

Consider a 1-dim parameter estimation problem, where the true parameter value is ϑ_0. Let $\hat{\vartheta}_{MVU}$ be a MVUE of ϑ_0. Consider the parametric set $\mathcal{F}$ of all estimators of the form $\hat{\vartheta}_1 = (1+a)\hat{\vartheta}_{MVU}$, $a \in \mathbb{R}$

(a) What can you infer from the fact that $\hat{\vartheta}_{MVU}$ is an unbiased estimator of ϑ_0?

(b) Prove that all $\hat{\vartheta}_1$'s of $\mathcal{F}$, $a \neq 0$, are biased estimators of ϑ_0.

(c) Find the $MSE(\hat{\vartheta}_{MVU})$. Explain why this value cannot be zero for finite N.

(d) Express the $MSE(\hat{\vartheta}_1)$ in terms of $MSE(\hat{\vartheta}_{MVU})$.

(e) Determine the range of values of "a" that result to estimators with MSE lower than $MSE(\hat{\vartheta}_{MVU})$.

(f) Prove that for any value of "a" in the range defined in (e), it's $|\vartheta_1| < |\hat{\vartheta}_{MVU}|$.

(g) Determine the value a^* of a that corresponds to the estimator giving the lowest MSE.

(h) Explain why in practice a^* cannot be determined.

(a) $E[\hat{\vartheta}_{MVU}] = \vartheta_0$

(b) $E[\vartheta_1] = E[(1+a)\hat{\vartheta}_{MVU}] = (1+a)E[\hat{\vartheta}_{MVU}] = (1+a)\vartheta_0$

for every $a \neq 0$, $E[\vartheta_1] \neq \vartheta_0 \Rightarrow \vartheta_1$ strictly biased.

(c) $MSE(\hat{\vartheta}_{MVU}) = E[(\hat{\vartheta}_{MVU} - E[\hat{\vartheta}_{MVU}])^2] + (E[\hat{\vartheta}_{MVU}] - \vartheta_0)^2 \overset{=0}{}$

$= E[(\hat{\vartheta}_{MVU} - E[\hat{\vartheta}_{MVU}])^2] \neq 0$ for finite N

(because, according to Cramer-Rao lower band, the variance of any unbiased estimator is banded by a positive value for finite N).

(d) $E[(\vartheta_1 - E[\vartheta_1])^2] = E[((1+a)\hat{\vartheta}_{MVU} - (1+a)\vartheta_0)^2]$

$= (1+a)^2 E[(\hat{\vartheta}_{MVU} - \vartheta_0)^2]$

$= (1+a)^2 \cdot MSE(\hat{\vartheta}_{MVU})$ $\Rightarrow$

$(E[\vartheta_1] - \vartheta_0)^2 = ((1+a)\vartheta_0 - \vartheta_0)^2 = a^2\vartheta_0^2$

$\Rightarrow MSE(\hat{\vartheta}_1) = (1+a)^2 MSE(\hat{\vartheta}_{MVU}) + a^2\vartheta_0^2$

(e) $MSE(\hat{\vartheta}_1) < MSE(\hat{\vartheta}_{MVU}) \overset{(d)}{\Rightarrow} (1+a)^2 MSE(\hat{\vartheta}_{MVU}) + a^2\vartheta_0^2 < MSE(\hat{\vartheta}_{MVU})$

$\Rightarrow a^2(MSE(\hat{\vartheta}_{MVU}) + \vartheta_0^2) + 2a \cdot MSE(\hat{\vartheta}_{MVU}) < 0$

$\Rightarrow a \cdot [a(MSE(\hat{\vartheta}_{MVU}) + \vartheta_0^2) + 2 \cdot MSE(\hat{\vartheta}_{MVU})] < 0$

$\Rightarrow \dfrac{-2MSE(\hat{\vartheta}_{MVU})}{MSE(\hat{\vartheta}_{MVU}) + \vartheta_0^2} < a < 0.$

(f) $MSE(\vartheta'_{MVU}) + \vartheta_0^2 > MSE(\vartheta'_{MVU}) \Rightarrow \frac{MSE(\vartheta'_{MVU})}{MSE(\vartheta'_{MVU}) + \vartheta_0^2} < 1$

$\Rightarrow \frac{-2\,MSE(\vartheta'_{MVU})}{MSE(\vartheta'_{MVU}) + \vartheta_0^2} > -2 \Rightarrow -2 < a < 0.$

$\Rightarrow -1 < a + 1 < 1 \Rightarrow |1 + a| < 1 \Rightarrow |\vartheta_3| < |\vartheta'_{MVU}|$

(g) $\frac{\partial\, MSE(\vartheta_3)}{\partial a} = 0 \Rightarrow 2(1+a)\,MSE(\vartheta'_{MVU}) + 2a\vartheta_0^2 = 0$

$\Rightarrow a^* = \frac{-MSE(\vartheta'_{MVU})}{MSE(\vartheta'_{MVU}) + \vartheta_0^2}$

(h) a^* is dependent on ϑ_0
ϑ_0 represents reality $\Rightarrow \vartheta_0$ practically unknown
$\Rightarrow a^*$ impossible to compute accurately. $\Rightarrow$

Ex. 14.

Consider a set N pairs (y_n, x_n), $n=1,\dots,N$, satisfying the equation

$$y_n = \theta_0^T x_n + \eta_n ,$$

where η_n is normally distributed zero mean i.i.d noise. As it is known, the LS estimator satisfies the equation

$$\left(\sum_{n=1}^{N} x_n x_n^T\right)\theta = \sum_{n=1}^{N} y_n x_n$$

Consider now the special case where the θ is a scalar and $x_n = 1$ for all n. In this case, $y_n = \theta_0 + \eta_n$.

Let y_n denote the rv that models y_n. Let also $\bar{y} = \frac{1}{N}\sum_{n=1}^{N} y_n$ be the mean of N statistically independent rv's that follow the same pdf. Each one of the y_n's models a y_n.

(a) Derive the LS estimator of θ_0 for this case.

(b) Prove that y_n is an unbiased estimator of θ_0.

(c) Prove that $\bar{y}$ is an unbiased estimator of θ_0.

(d) Since $\bar{y}$ is the LS estimator for the 1-dim case, it's known that it's minimum variance unbiased estimator. From now on it will be denoted by $\hat{\theta}_{mvu}$.

(e) Prove that the ridge regression estimator for the present 1-dim case is expressed as $\hat{\theta} = \frac{\sum_{n=1}^{N} y_n}{N+1}$

(f) Denoting as $\hat{\theta}$ the ridge regression estimator, express it in terms of $\hat{\theta}_{mvu}$.

(g) Prove that $\hat{\theta}$ is a biased estimator.

(h) Verify that $|\hat{\theta}| < |\hat{\theta}_{mvu}|$.

(a) for $x_n = 1$ $\forall n$: $\left(\sum_{n=1}^{N} 1\cdot 1\right)\theta = \sum_{n=1}^{N} y_n \cdot 1$

$$\Rightarrow N\theta = \sum_{n=1}^{N} y_n$$

$$\Rightarrow \hat{\theta} = \frac{1}{N}\sum_{n=1}^{N} y_n = \bar{y}$$

(b) $E[y_n] = E[\theta_0 + \eta_n] = \theta_0 + \overbrace{E[\eta_n]}^{0} = \theta_0$

(c) $E[\bar{y}] = \frac{1}{N}\sum \theta_0 = \frac{1}{N}\cdot N\cdot\theta_0 = \theta_0$

(e) The formula proved in ex. 1a: $\left(\sum_{n=1}^{N} x_n x_n^T + \lambda I\right)\hat{\theta} = \sum_{n=1}^{N} y_n x_n$

for $x_n = 1$ and $I = 1$ (1-dim problem):

$$\left(\sum_{n=1}^{N} 1 + \lambda\right)\hat{\theta} = \sum_{n=1}^{N} y_n \Rightarrow \hat{\theta} = \frac{\sum_{n=1}^{N} y_n}{N+\lambda}$$

(f) $$\sum_{n=1}^{N} y_n = N \cdot \bar{y} = N \cdot \vartheta'_{MVU}$$

$$\hat{\vartheta} = \frac{\sum_{n=1}^{N} y_n}{N+\lambda} = \frac{N \cdot \vartheta'_{MVU}}{N+\lambda}$$

(g) $$E(\hat{\vartheta}) = E\left(\frac{N}{N+\lambda} \cdot \vartheta'_{MVU}\right) = \frac{N}{N+\lambda} \cdot E(\vartheta'_{MVU})$$

$$E(\vartheta'_{MVU}) = \vartheta_0 \Rightarrow$$

$$\Rightarrow E(\hat{\vartheta}) = \frac{N}{N+\lambda} \cdot \vartheta_0$$

$$\lambda > 0 \Rightarrow \frac{N}{N+\lambda} \neq 1 \Rightarrow E(\hat{\vartheta}) \neq \vartheta_0$$

(h) $$|\hat{\theta}| = \left|\frac{N}{N+\lambda}\right| \cdot |\vartheta'_{MVU}| \Rightarrow \frac{|\hat{\theta}|}{|\vartheta'_{MVU}|} = \left|\frac{N}{N+\lambda}\right| \quad (1)$$

$$N, \lambda > 0 \Rightarrow \frac{N}{N+\lambda} < 1 \Rightarrow |\hat{\vartheta}| < |\vartheta'_{MVU}|$$